



# SAFETY DATA SHEET

This safety data sheet complies with the requirements of:  
SS586: 2008 (2014)

Revision date 16-Oct-2024

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** ReadyPrep 2-D Rehydration/Sample Buffer

### Other means of identification

**Catalogue Number(s)** 1632106, 9704001

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

For further information, please contact

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Singapore) PTE LTD  
3A International Business Park #11-10/16  
ICON@IBP  
Singapore 609935

#### **Technical Service**

6424 0262  
ctssingapore@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Singapore: 65-31581349

## SECTION 2: Hazards identification

### GHS Classification

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS): SS586: 2008 (2014)

### Label elements

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS): SS586: 2008 (2014)

### Other hazards which do not result in classification

No information available.

## SECTION 3: Composition/information on ingredients

**Substance**

Not applicable

**Mixture**

| Chemical name                                                                                                      | EC No (EU Index No)         | CAS No.    | Weight-%     |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------|------------|--------------|
| Urea                                                                                                               | 200-315-5                   | 57-13-6    | 50 - 100     |
| CHAPS                                                                                                              | -                           | 75621-03-3 | 2.5 - 5      |
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)-                                                                          | 222-468-7                   | 3483-12-3  | 1 - 2.5      |
| Water                                                                                                              | 231-791-2                   | 7732-18-5  | 0.3 - 0.99   |
| 2-Propenamide, N,N-methylenebis-, polymer with 2-propenamide                                                       | -                           | 25034-58-6 | 0.001 - 0.01 |
| 1,2,3-Propanetriol                                                                                                 | 200-289-5                   | 56-81-5    | 0.001 - 0.01 |
| Phenol, 4,4-(1,1-dioxido-3H-2,1-benzoxathiol-3-ylidene)bis[2,6-dibromo-3,6,9,12-Tetrazaatetradecane-1,14-diamine   | 204-086-2                   | 115-39-9   | 0.001 - 0.01 |
| Ethyl acrylate                                                                                                     | 223-775-9<br>(612-064-00-2) | 4067-16-7  | 0.001 - 0.01 |
| Methanamine, N-methyl-, reaction products with chloromethylated divinylbenzene-ethenylethylbenzene-styrene polymer | 205-438-8<br>(607-032-00-X) | 140-88-5   | < 0.001      |
| DL-Lysine, monohydrochloride                                                                                       | -                           | 69011-17-2 | < 0.001      |
| Sodium azide                                                                                                       | 200-739-0                   | 70-53-1    | < 0.001      |
| L-Arginine, monohydrochloride                                                                                      | 247-852-1<br>(011-004-00-7) | 26628-22-8 | < 0.001      |
| N-Glycyl-L-glutamic acid                                                                                           | 214-275-1                   | 1119-34-2  | < 0.001      |
| Aspartic acid, glycyl-                                                                                             | 231-019-4                   | 7412-78-4  | < 0.001      |
| DL-Glutamic acid                                                                                                   | -                           | 79731-35-4 | < 0.001      |
| DL-Aspartic acid                                                                                                   | 210-522-2                   | 617-65-2   | < 0.001      |
|                                                                                                                    | 210-513-3                   | 617-45-8   | < 0.001      |

**SECTION 4: First aid measures****Description of first aid measures**

|                                                                    |                                                                                                                   |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                                              | No hazards which require special first aid measures.                                                              |
| <b>Inhalation</b>                                                  | Remove to fresh air.                                                                                              |
| <b>Eye contact</b>                                                 | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| <b>Skin contact</b>                                                | Wash skin with soap and water. In the case of skin irritation or allergic reactions see a doctor.                 |
| <b>Ingestion</b>                                                   | Rinse mouth thoroughly with water.                                                                                |
| <b>Most important symptoms and effects, both acute and delayed</b> |                                                                                                                   |
| <b>Symptoms</b>                                                    | Prolonged contact may cause redness and irritation.                                                               |

For emergency responders

Self-protection of the first aider      No information available.

Indication of any immediate medical attention and special treatment needed

Note to doctors      Treat symptomatically.

**SECTION 5: Firefighting measures**Suitable Extinguishing Media

Suitable Extinguishing Media      Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media      No information available.

Specific hazards arising from the chemical

Specific hazards arising from the chemical      None known.

Special protective actions for fire-fighters

Special protective equipment and precautions for fire-fighters      Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.  
Use personal protection equipment.

**SECTION 6: Accidental release measures**Personal precautions, protective equipment and emergency procedures

Personal precautions      Ensure adequate ventilation.

For emergency responders      Use personal protection recommended in Section 8.

Environmental precautions

Environmental precautions      See Section 12 for additional Ecological Information.

Methods and material for containment and cleaning up

Methods for containment      Prevent further leakage or spillage if safe to do so.

Methods for cleaning up      Pick up and transfer to properly labelled containers.

Prevention of secondary hazards      Clean contaminated objects and areas thoroughly observing environmental regulations.

Reference to other sections      See section 8 for more information. See section 13 for more information.

**SECTION 7: Handling and storage**Precautions for safe handling

Advice on safe handling      Ensure adequate ventilation.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep container tightly closed in a dry and well-ventilated place.

## SECTION 8: Exposure controls/personal protection

**Control parameters**

**Occupational exposure limits**

| Chemical name                 | Singapore                                                                             | ACGIH TLV                                                                              |
|-------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | PEL: 10 mg/m <sup>3</sup>                                                             | No data available                                                                      |
| Ethyl acrylate<br>140-88-5    | PEL: 5 ppm<br>PEL: 20 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 61 mg/m <sup>3</sup> | STEL: 15 ppm<br>TWA: 5 ppm                                                             |
| Sodium azide<br>26628-22-8    | STEL: 0.29 mg/m <sup>3</sup><br>STEL: 0.11 ppm                                        | Ceiling: 0.29 mg/m <sup>3</sup> Sodium azide<br>Ceiling: 0.11 ppm Hydrazoic acid vapor |

**Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Skin and body protection** Wear suitable protective clothing.

**Hand protection** Wear suitable gloves.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

## SECTION 9: Physical and chemical properties

**Information on basic physical and chemical properties**

**Physical state** Solid  
**Appearance** crystalline  
**Colour** white  
**Odour** Odourless.  
**Odour threshold** No information available

| <u>Property</u>                | <u>Values</u>     | <u>Remarks • Method</u> |
|--------------------------------|-------------------|-------------------------|
| pH                             |                   | None known              |
| Melting point / freezing point | No data available | None known              |

|                                         |                          |            |
|-----------------------------------------|--------------------------|------------|
| Initial boiling point and boiling range | No data available        | None known |
| Flash point                             | No data available        | None known |
| Evaporation rate                        | No data available        | None known |
| Flammability                            | No data available        | None known |
| Flammability Limit in Air               |                          | None known |
| Upper flammability or explosive limits  | No data available        |            |
| Lower flammability or explosive limits  | No data available        |            |
| Vapour pressure                         | No data available        | None known |
| Relative vapour density                 | No data available        | None known |
| Relative density                        | No data available        | None known |
| Water solubility                        | Soluble in water         |            |
| Solubility(ies)                         | No data available        | None known |
| Partition coefficient                   | No data available        | None known |
| Autoignition temperature                | No data available        | None known |
| Decomposition temperature               |                          | None known |
| Kinematic viscosity                     | No data available        | None known |
| Dynamic viscosity                       | No data available        | None known |
| <u>Other information</u>                | No information available |            |

## SECTION 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

Possibility of hazardous reactions None under normal processing.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials None known based on information supplied.

### Hazardous decomposition products

Hazardous decomposition products None known based on information supplied.

## SECTION 11: Toxicological information

### Information on likely routes of exposure

#### Product Information

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Inhalation Specific test data for the substance or mixture is not available.

|                     |                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------|
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available.                              |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Causes mild skin irritation. |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available.                              |

**Symptoms related to the physical, chemical and toxicological characteristics**

|                 |                                                     |
|-----------------|-----------------------------------------------------|
| <b>Symptoms</b> | Prolonged contact may cause redness and irritation. |
|-----------------|-----------------------------------------------------|

**Acute toxicity****Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 7,072.70 mg/kg

**Component Information**

| Chemical name                             | Oral LD50             | Dermal LD50             | Inhalation LC50               |
|-------------------------------------------|-----------------------|-------------------------|-------------------------------|
| Urea                                      | = 8471 mg/kg ( Rat )  |                         |                               |
| Water                                     | > 90 mL/kg ( Rat )    |                         |                               |
| 1,2,3-Propanetriol                        | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit )    | > 2.75 mg/L ( Rat ) 4 h       |
| 3,6,9,12-Tetraazatetradecane-1,14-diamine | = 1600 mg/kg ( Rat )  |                         |                               |
| Ethyl acrylate                            | = 550 mg/kg ( Rat )   | = 1790 mg/kg ( Rabbit ) | = 1410 ppm ( Rat ) 4 h        |
| Sodium azide                              | = 27 mg/kg ( Rat )    | = 20 mg/kg ( Rabbit )   | 0.054 - 0.52 mg/L ( Rat ) 4 h |
| L-Arginine, monohydrochloride             | = 12 g/kg ( Rat )     |                         |                               |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                      |
|------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Causes mild skin irritation. |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met.                    |
| <b>Respiratory or skin sensitisation</b> | Based on available data, the classification criteria are not met.                    |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met.                    |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met.                    |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met.                    |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met.                    |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met.                    |
| <b>Aspiration hazard</b>                 | Classification not possible.                                                         |

**SECTION 12: Ecological information**

**Ecotoxicity****Ecotoxicity**

| Chemical name      | Algae/aquatic plants                                | Fish                                                                                                                                                         | Crustacea                                    |
|--------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Urea               | -                                                   | LC50: 16200 - 18300mg/L (96h, <i>Poecilia reticulata</i> )                                                                                                   | EC50: =3910mg/L (48h, <i>Daphnia magna</i> ) |
| 1,2,3-Propanetriol | -                                                   | LC50: 51 - 57mL/L (96h, <i>Oncorhynchus mykiss</i> )                                                                                                         | -                                            |
| Ethyl acrylate     | EC50: =48mg/L (72h, <i>Desmodemus subspicatus</i> ) | LC50: =4.6mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: 2.31 - 2.7mg/L (96h, <i>Pimephales promelas</i> )                                                 | EC50: =7.9mg/L (48h, <i>Daphnia magna</i> )  |
| Sodium azide       | -                                                   | LC50: =0.8mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: =0.7mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =5.46mg/L (96h, <i>Pimephales promelas</i> ) | -                                            |

**Persistence and degradability**

**Persistence and degradability** No information available.

**Bioaccumulative potential**

**Bioaccumulation** No information available.

| Chemical name      | Partition coefficient |
|--------------------|-----------------------|
| Urea               | -1.73                 |
| 1,2,3-Propanetriol | -1.75                 |
| Ethyl acrylate     | 1.18                  |

**Mobility**

**Mobility in soil** No information available.

**PBT and vPvB assessment**

| Chemical name                 | PBT and vPvB assessment         |
|-------------------------------|---------------------------------|
| Urea                          | The substance is not PBT / vPvB |
| 1,2,3-Propanetriol            | The substance is not PBT / vPvB |
| Ethyl acrylate                | The substance is not PBT / vPvB |
| Sodium azide                  | The substance is not PBT / vPvB |
| L-Arginine, monohydrochloride | The substance is not PBT / vPvB |

**Other adverse effects**

**Other adverse effects** No information available

**SECTION 13: Disposal considerations****Disposal methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging

Do not reuse empty containers.

**SECTION 14: Transport information****IMDG**

Not regulated

Transport in bulk according to  
Annex II of MARPOL and the IBC  
Code

No information available

**IATA**

Not regulated

**SECTION 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****Singapore****Environmental Protection and Management (Hazardous Substances) Regulations**

Verify that licence requirements are met.

| Chemical name | Hazardous Substances                          | transport |
|---------------|-----------------------------------------------|-----------|
| Sodium azide  | Exclusions: Air bag devices in motor vehicles | 0kg       |

**Environmental Public Health Act**

Dispose of waste product or used containers according to local regulations.

**Fire Safety (Petroleum and Flammable Materials) Regulations**

Verify that licence requirements are met.

| Chemical name  | Regulated    | Hazard class |
|----------------|--------------|--------------|
| Ethyl acrylate | SCDEAR1917L2 | 3            |

**Hazardous Waste (Control of Export, Import and Transit) Act**

It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations. Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

**Poison**

Verify that licence requirements are met Verify that requirements related to using, handling, and storing substances subject to prohibition, authorisation or restriction are met

| Chemical name | Poison | Poison Schedule Number |
|---------------|--------|------------------------|
| Urea          | X      | First schedule         |

**Workplace Safety and Health Act**

See section 8 for national exposure control parameters. Comply with the health and safety at work laws.

**International Regulations****The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable**The Stockholm Convention on Persistent Organic Pollutants** Not applicable**The Rotterdam Convention** Not applicable



**International Inventories**

Contact supplier for inventory compliance status

**SECTION 16: Other information****Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Label elements**

|                      |                                                          |
|----------------------|----------------------------------------------------------|
| <b>Issuing Date</b>  | Bio-Rad Laboratories, Environmental Health and Safety    |
| <b>Revision date</b> | 16-Oct-2024                                              |
| <b>Revision Note</b> | Significant changes throughout SDS. Review all sections. |

**This safety data sheet complies with the requirements of: SS586: 2008 (2014)**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**